الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

**Faculty of Artificial Intelligence Engineering**

Department of Software Engineering
and Intelligent Information Systems

**QARIB**

# Community Engagement Platform

Senior 2 Project

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# Abstract

Qarib is a proposed community engagement platform that connects citizens with responsible organizations through a unified digital environment. It addresses the fragmentation of complaint submission, limited visibility into complaint progress, and the separation between public initiatives and citizen feedback. The platform brings these activities together while assigning clear responsibilities to citizens, organization representatives, and system administrators.

Citizens can submit complaints with a description, a selected category, a location, and a supporting photo. Each complaint receives a unique identifier and is routed according to the selected category. Citizens can track status changes, review processing notes and resolution evidence, and receive notifications. Organization representatives manage complaints within their assigned categories and manage initiatives within their organizations. Citizens can browse initiatives, filter them by date or location, submit written feedback, and view initiative statistics. Administrators manage accounts and review aggregate platform dashboards.

The planned AI functionality checks complaint text for abusive language and generates an automatic acknowledgment after successful submission. A first abusive submission results in a warning, while repeated abuse triggers a 14-day account suspension. Complaint routing relies on the citizen-selected category rather than AI classification. Geographic heatmaps and charts support the presentation of complaint and initiative data.

The project follows a Scrum-based development plan organized around five functional iterations. This document presents the project background, objectives, scope, methodology, reference study, requirements, and approved behavioral models. The expected outcome is a working platform that supports more transparent follow-up, organized service coordination, and informed community participation. These benefits are project objectives to be evaluated through implementation and testing.

Keywords: community engagement; complaints; government initiatives; written feedback; geographic heatmaps; Scrum.

# الملخص

قريب هو منصة مقترحة للمشاركة المجتمعية تربط المواطنين بالجهات المسؤولة ضمن بيئة رقمية موحدة. يعالج المشروع تشتت قنوات تقديم الشكاوى، وضعف وضوح مراحل معالجتها، وانفصال المعلومات المتعلقة بالمبادرات العامة عن آراء المواطنين. وتجمع المنصة هذه الأنشطة مع تحديد مسؤوليات واضحة لكل من المواطن وممثل الجهة ومدير النظام.

يستطيع المواطن تقديم شكوى تتضمن وصفاً وفئة يختارها وموقعاً وصورة داعمة. وتحصل كل شكوى على رقم فريد، ثم توجه إلى الممثل المسؤول بحسب الفئة المحددة. كما يستطيع المواطن متابعة تغيرات الحالة والاطلاع على ملاحظات المعالجة وصورة إثبات الحل وتلقي الإشعارات. ويتولى ممثلو الجهات معالجة الشكاوى ضمن الفئات المسندة إليهم وإدارة مبادرات جهاتهم. ويمكن للمواطنين تصفح المبادرات وتصفيتها بحسب التاريخ أو الموقع وإرسال ملاحظات مكتوبة والاطلاع على إحصائيات المبادرات، بينما يدير مسؤول النظام الحسابات ويطّلع على الإحصائيات الإجمالية للمنصة.

تشمل وظائف الذكاء الاصطناعي المخطط لها فحص نص الشكوى للكشف عن الألفاظ المسيئة، وتوليد إشعار تلقائي يؤكد استلام الشكوى بعد نجاح تقديمها. تؤدي المخالفة الأولى إلى تحذير، ويؤدي تكرار الإساءة إلى إيقاف الحساب لمدة أربعة عشر يوماً. ويعتمد توجيه الشكوى على الفئة التي يختارها المواطن، وليس على تصنيف آلي. وتستخدم الخرائط الحرارية والمخططات البيانية لعرض البيانات المكانية والإحصائية للشكاوى والمبادرات.

يتبع المشروع خطة تطوير مبنية على سكرم، ومقسمة إلى خمس دورات وظيفية. وتعرض هذه الوثيقة خلفية المشروع وأهدافه ونطاقه ومنهجية تطويره والدراسة المرجعية والمتطلبات والنماذج السلوكية المعتمدة. والنتيجة المتوقعة هي منصة عملية تدعم شفافية المتابعة وتنظيم التنسيق بين المواطنين والجهات المعنية وتعزيز المشاركة المجتمعية المستندة إلى المعلومات. وتعد هذه الفوائد أهدافاً للمشروع يجري تقييمها خلال التنفيذ والاختبار.

الكلمات المفتاحية: المشاركة المجتمعية، الشكاوى، المبادرات الحكومية، الملاحظات المكتوبة، الخرائط الحرارية، سكرم.

# Chapter

# 01

# Project Overview and Background

## 1.1 Project Overview

Qarib is a community engagement platform intended to provide one place for citizens to report local service problems, follow their complaints, explore government initiatives, and provide written feedback. It organizes communication through three roles: Citizen, Organization Representative, and System Admin. The proposed application is designed for web access, with a responsive interface suitable for everyday use.

## 1.2 Background and Context

Local service concerns, such as damaged roads, waste accumulation, water issues, or lighting faults, require a clear route from the person observing the problem to the responsible organization. Public initiatives also need an accessible channel through which citizens can discover information and share their views. Qarib brings complaint follow-up and initiative engagement into a common workflow. The reference study in Chapter 3 examines related platforms and compares their selected capabilities with the proposed system.

## 1.3 Problem Statement

The project addresses a workflow in which complaints and initiative information may be distributed across separate communication channels. Without a structured record, a citizen may not have a reliable reference for following a complaint, while an organization may lack a consistent way to record responsibility, processing notes, and resolution evidence. Citizen feedback on initiatives may also be difficult to connect with the initiative concerned.

## 1.4 Project Significance

A unified platform can improve the visibility of complaint progress and provide a traceable record of actions. Category-based assignment clarifies responsibility, while processing notes and resolution photos support follow-up. Initiative pages and written feedback provide a structured participation channel. Aggregate dashboards and geographic views can help users identify concentrations and patterns. These are intended benefits; the project does not assume that response times or service quality have already improved.

# Project Objectives

## 1.5 Main Objective

To design and develop a community engagement platform that integrates complaint submission and follow-up, organization initiative management, citizen feedback, and role-based statistics within a consistent and traceable workflow.

## 1.6 Specific Objectives

**1. Provide structured complaint submission.** Capture the description, citizen-selected category, location, and supporting photo, and return a unique complaint identifier.

**2. Support transparent follow-up.** Allow citizens to view and search their own complaints, inspect processing notes and resolution photos, and receive status notifications.

**3. Organize complaint responsibility.** Route complaints using the configured category mapping, support corrections with a recorded reason, and preserve complaints as Pending Assignment when no responsible representative is configured.

**4. Enable organization-level initiative management.** Allow representatives to create, update, and delete their organization's initiatives and view related written feedback.

**5. Support informed participation.** Let citizens browse and filter initiatives, view details, submit written feedback, and explore initiative statistics.

**6. Provide administrative control and insight.** Support account creation, updates, activation and deactivation, and dashboards covering complaints, initiatives, feedback summaries, and geographic distributions.

**7. Integrate focused AI assistance.** Check submitted complaint text for abusive language and automatically acknowledge successful submissions, following the agreed warning and suspension rules.

**8. Verify the implemented workflows.** Use the approved use cases, sequence diagrams, and selected activity diagrams to guide implementation, functional testing, and integration checks.

# Project Scope

## 1.7 Functional Scope

**Citizen:** account access and profile management; complaint submission, search and tracking; access to notes, resolution evidence and status notifications; initiative browsing, filtering and written feedback; and public initiative statistics.

**Organization Representative:** personal profile management with a read-only organization name; processing complaints in the assigned category; correcting complaint assignment with a reason; and managing initiatives and viewing feedback within the assigned organization.

**System Admin:** account management for all supported roles, including the admin's own account; organization and complaint-category assignment for representatives; and aggregate platform dashboards.

## 1.8 Scope Rules and Boundaries

Complaint responsibility is determined by the citizen-selected category, regardless of location. If no representative is mapped to that category, the complaint is retained as Pending Assignment. Representatives may update a complaint to In Progress, Resolved, or Rejected; a resolution photo is required when the complaint is resolved.

Initiative feedback is written feedback. Star ratings, proof of attendance, and a requirement that an initiative has ended are not part of the approved feedback use case. Administrators manage accounts and view aggregate information; direct complaint processing and initiative management belong to organization representatives.

The AI scope is limited to abuse checking and automatic receipt acknowledgment. The first abusive submission triggers a warning; repeated abuse triggers a 14-day suspension. Automatic AI category classification and an open-ended citizen-assistant conversation are outside the approved use cases.

## 1.9 Deliverables and Evaluation Boundary

Deliverables comprise the analysis and design documentation, the planned application implementation, role-based data storage and access, functional and integration checks, and final code review. The system records and communicates complaint handling; it does not perform physical maintenance or guarantee that an external organization will resolve a complaint within a specified time.

# Chapter

# 02

# Development Methodology

## 2.1 Selected Approach: Scrum

Scrum organizes work into short Sprints and uses inspection and adaptation to guide progress. A Product Backlog orders the required work; Sprint Planning selects a goal and work for the iteration. Daily Scrum, Sprint Review, and Sprint Retrospective support progress inspection and improvement. Each Sprint aims to produce a usable Increment that meets the Definition of Done. The Scrum Guide describes Sprints as fixed-length periods of one month or less. [1]

## 2.2 Reasons for Selecting Scrum

Qarib contains distinct but connected modules: accounts, complaints, initiatives, dashboards, and AI assistance. Incremental development makes it possible to implement and check these modules in stages. Regular reviews provide opportunities to clarify workflow details and incorporate academic feedback before later modules depend on them.

The project also includes implementation uncertainty, particularly in AI integration and geographic visualization. A prioritized backlog allows technical discoveries and defects to be addressed during development. Testing and integration inside each iteration help reveal compatibility problems earlier, rather than postponing all verification until the end.

## 2.3 Application to This Academic Project

The project will use a Scrum-based academic adaptation for an individual developer. Shahed Alzoni will maintain the backlog, plan implementation work, record progress, and prepare reviewable increments. Academic supervisors will provide feedback at agreed checkpoints; this document does not assign them formal Scrum roles.

The supplied Gantt chart contains 17-day and 16-day iterations and an initial analysis period. These are the approved project planning durations. Because the plan is individual and the iteration durations vary, it is described as a Scrum-based adaptation rather than a complete implementation of the standard Scrum team structure.

[1] Schwaber, K., and Sutherland, J. (2020). The Scrum Guide. https://scrumguides.org/scrum-guide.html

# Iteration Plan and Quality Checks

## 2.4 Planned Development Workflow

At the start of each iteration, the relevant use cases will be translated into implementation tasks and acceptance checks. A short daily progress review will record completed work, the next tasks, and blockers. At the end, a working increment will be reviewed and remaining issues returned to the backlog. A brief retrospective will identify practical improvements for the following iteration.

An item will be treated as complete when its planned behavior is implemented, access restrictions and relevant error paths have been checked, and it works with the modules it depends on. The Gantt chart includes detailed design, backend implementation, frontend implementation, and testing and integration within each functional iteration.

## 2.5 Schedule Overview

The supplied plan runs from **03 October 2026 to 23 January 2027**. Scope determination is followed by project planning and system analysis, with initial analysis ending on 31 October 2026. Five functional iterations follow, and the plan ends with code review.

| Iteration | Functional focus | Planned duration |
|---|---|---|
| Sprint 1 | Account Management | 17 days |
| Sprint 2 | Complaint Management | 17 days |
| Sprint 3 | Initiative Management | 17 days |
| Sprint 4 | Platform Dashboard | 16 days |
| Sprint 5 | AI Virtual Assistant | 16 days |

The final AI iteration covers the approved abuse-checking and receipt-acknowledgment functionality. Its integration will be checked against the complaint workflow implemented earlier. The following page reproduces the supplied Gantt chart, including its task durations, sequence, and assigned resource.

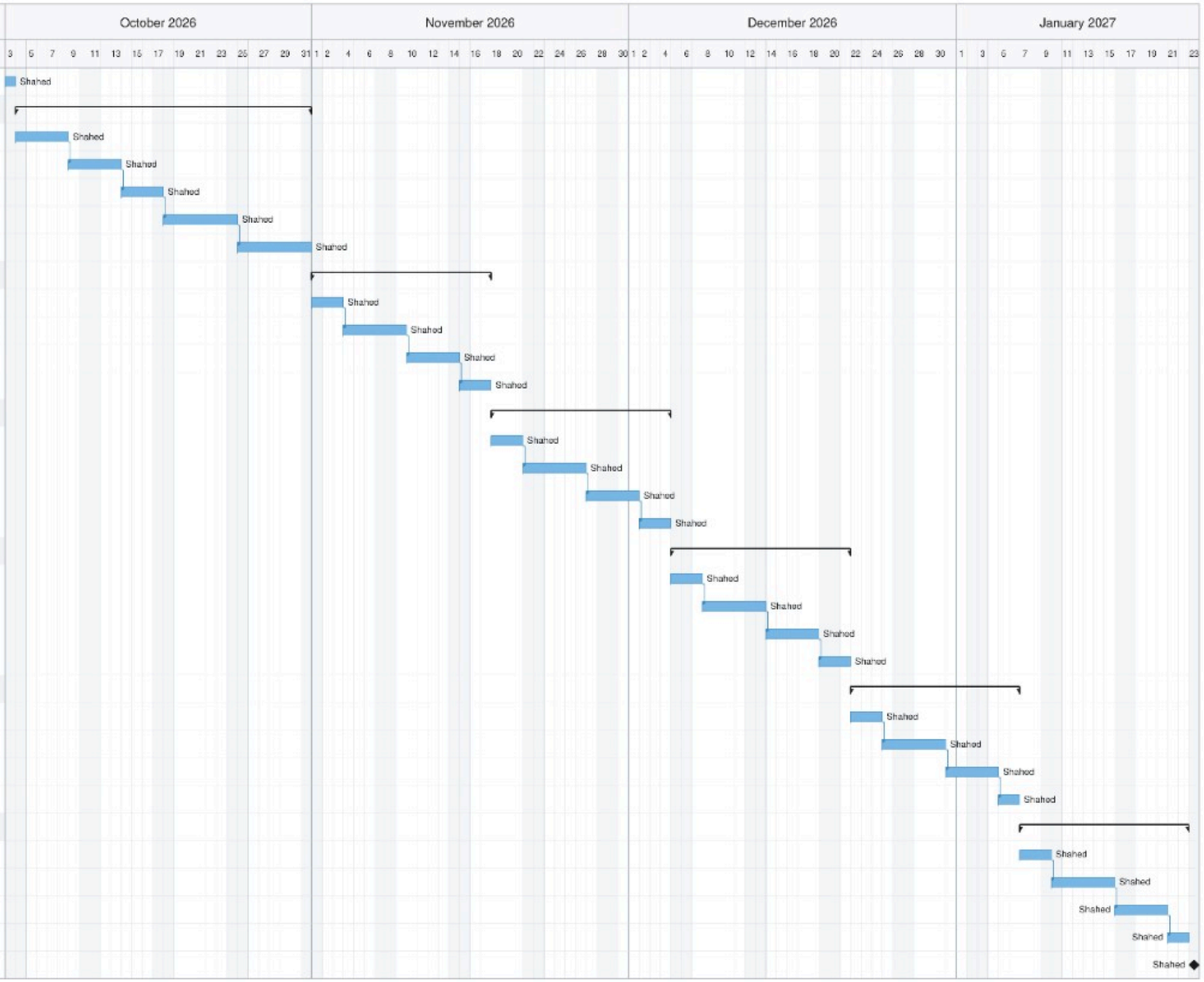
Schedule source: project Gantt chart supplied by the student. Durations and task names are preserved as shown in the chart.

## 2.6 Project Gantt Chart

**Qarib - Community Engagement Platform**

Scrum Gantt Chart | 03 Oct 2026 - 23 Jan 2027 | Assigned To: Shahed

| ID | Task Mode | Task Name | Duration |
|---|---|---|---|
| 1 | | Scope Determination | 1 day |
| 2 | | **Project Planning & System Analysis** | **28 days** |
| 3 | | Literature Review | 5 days |
| 4 | | Product Backlog | 5 days |
| 5 | | Use Case Diagram | 4 days |
| 6 | | Use Case Specifications | 7 days |
| 7 | | System Architecture & High-Level Design | 7 days |
| 8 | | **Sprint 1: Account Management** | **17 days** |
| 9 | | Detailed Design | 3 days |
| 10 | | Backend Implementation | 6 days |
| 11 | | Frontend Implementation | 5 days |
| 12 | | Testing & Integration | 3 days |
| 13 | | **Sprint 2: Complaint Management** | **17 days** |
| 14 | | Detailed Design | 3 days |
| 15 | | Backend Implementation | 6 days |
| 16 | | Frontend Implementation | 5 days |
| 17 | | Testing & Integration | 3 days |
| 18 | | **Sprint 3: Initiative Management** | **17 days** |
| 19 | | Detailed Design | 3 days |
| 20 | | Backend Implementation | 6 days |
| 21 | | Frontend Implementation | 5 days |
| 22 | | Testing & Integration | 3 days |
| 23 | | **Sprint 4: Platform Dashboard** | **16 days** |
| 24 | | Detailed Design | 3 days |
| 25 | | Backend Implementation | 6 days |
| 26 | | Frontend Implementation | 5 days |
| 27 | | Testing & Integration | 2 days |
| 28 | | **Sprint 5: AI Virtual Assistant** | **16 days** |
| 29 | | Detailed Design | 3 days |
| 30 | | Backend Implementation | 6 days |
| 31 | | Frontend Implementation | 5 days |
| 32 | | Testing & Integration | 2 days |
| 33 | | **Code Review** | 1 day |




Project: Qarib
Start: 03/10/2026
Finish: 23/01/2027
Resource: Shahed
Task
Summary
Code Review
*Planned calendar-day durations. Initial analysis ends on 31 Oct 2026.*

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# Chapter

# 03

# OneService

REFERENCE STUDY

**Country** **Singapore**

**Link** https://www.oneservice.gov.sg/oneservice/

## Overview

OneService is a government municipal-service channel that allows residents to submit feedback about local issues. Reports are routed to the relevant agencies and town councils, and residents can follow case updates.

## Interface examples




Interface examples showing issue categories, report subcategories, and map-based location selection.

# MUNIPOLIS



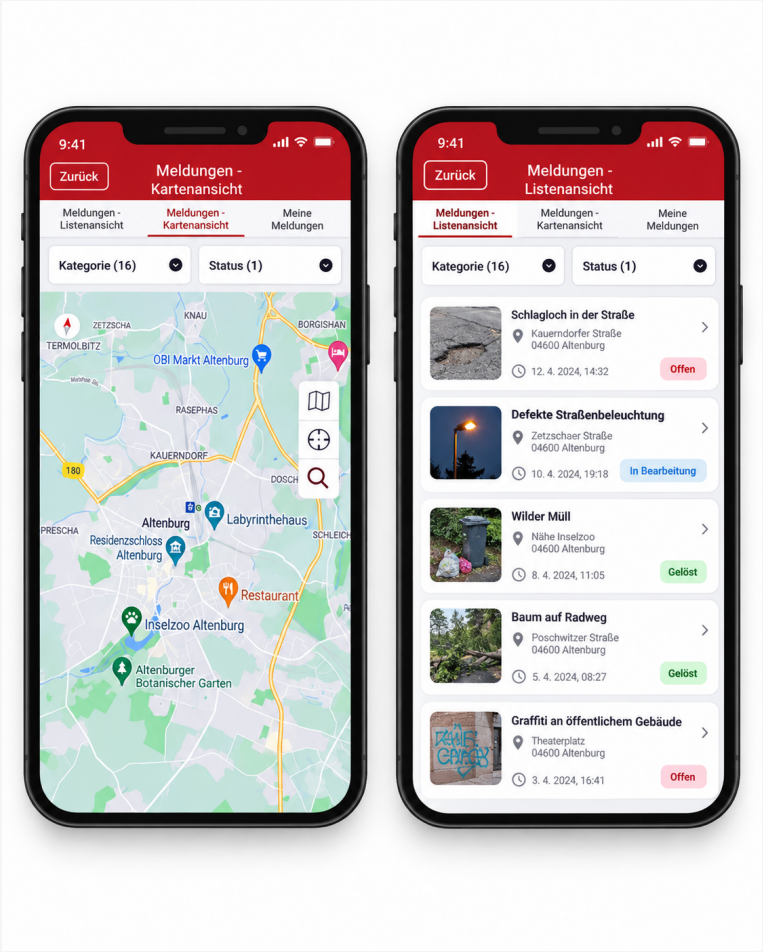
**Country** **Czechia**

**Link** https://info.munipolis.com/

## Overview

MUNIPOLIS is a municipal communication platform that distributes official local information and supports citizen feedback. Its services include local notices, events, targeted alerts, surveys, and issue reporting.

## Interface examples




Example of the map interface used to view reported issues and their locations.

# Dubai Municipality App

REFERENCE STUDY

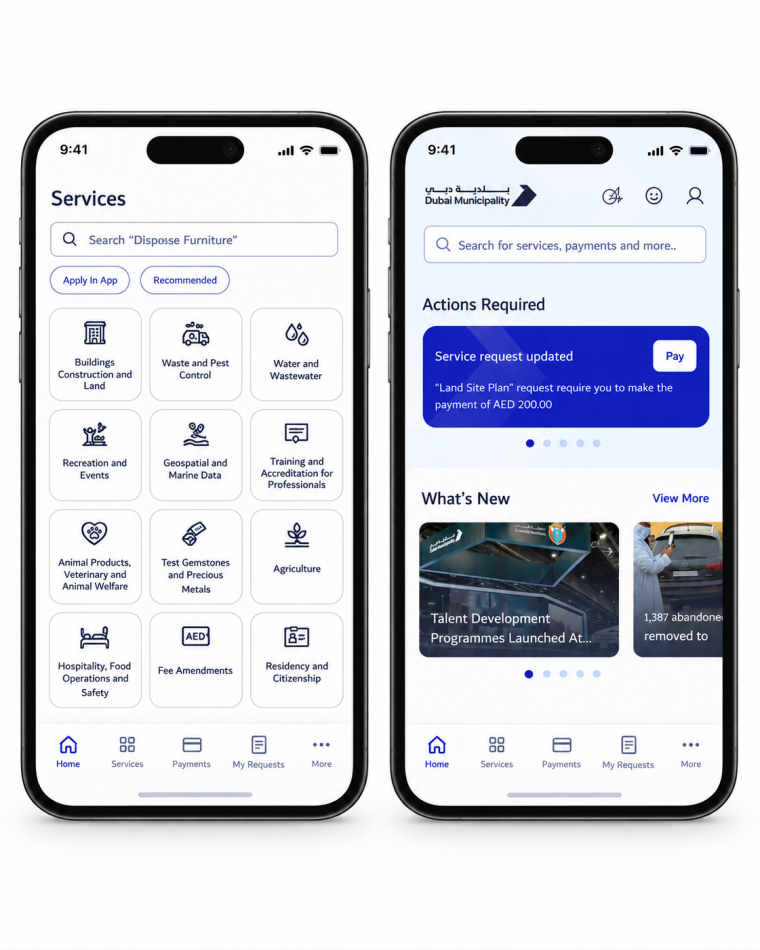
**Country** **United Arab Emirates**

**Link** https://www.dm.gov.ae/dubai-municipality-app/

## Overview

The Dubai Municipality App provides a unified digital gateway to municipal services and city information. Citizens can submit requests or complaints, receive notifications, track request status, and provide feedback.

## Interface examples




Interface examples showing municipal service categories, search, required actions, and service updates.

# Comparative Reference Study



The table compares selected services with the functionality of the proposed platform.

| Capability | OneService | MUNIPOLIS | Dubai App | Our System |
|---|---|---|---|---|
| Citizen complaint submission | ✓ | ✓ | ✓ | ✓ |
| Photo attachment | ✓ | ✓ | ✗ | ✓ |
| Map-based complaint location | ✓ | ✓ | ✓ | ✓ |
| Complaint category routing | ✓ | ✗ | ✗ | ✓ |
| Complaint status tracking | ✓ | ✓ | ✓ | ✓ |
| Government initiatives | ✗ | ✓ | ✗ | ✓ |
| Initiative filtering | ✗ | ✗ | ✗ | ✓ |
| Public initiative dashboard | ✗ | ✗ | ✗ | ✓ |
| Initiative written feedback | ✗ | ✗ | ✗ | ✓ |
| Citizen notifications | ✓ | ✓ | ✓ | ✓ |
| Representative complaint workspace | ✓ | ✓ | ✓ | ✓ |
| Aggregate platform statistics | ✗ | ✓ | ✓ | ✓ |
| Complaint heatmap | ✗ | ✗ | ✗ | ✓ |
| Responsive web access | ✗ | ✓ | ✗ | ✓ |

Legend: check = included / available; cross = not included / not identified in the reference study.

# Chapter

# 04

# Actor 1 - Citizen



## ACCOUNT AND PROFILE

1. Register using an email address and mobile number.
2. Sign in and sign out.
3. Recover a forgotten password.
4. View and update personal profile information.
5. Deactivate the personal account.

## COMPLAINTS

6. Submit a complaint with a description, selected category, location, and supporting photo.
7. Receive a unique complaint ID and an automatic AI-generated acknowledgment after submission.
8. View personal complaints and track their statuses.
9. Search for a personal complaint using its complaint ID.
10. View processing notes and resolution evidence.
11. Receive notifications when a complaint status changes.

## GOVERNMENT INITIATIVES

12. Browse government initiatives, including projects and public events.
13. Filter initiatives by date and location.
14. View initiative details and submit written feedback.

## INITIATIVE STATISTICS

15. View initiative activity on a geographic heatmap.
16. View charts of initiative counts by area and organizing organization.

## AUTOMATED SYSTEM BEHAVIOR

- The AI checks submitted complaint text for abusive language and issues a first warning.
- A repeated abusive submission triggers a 14-day account suspension.

# Actor 2 - Organization Representative



### ACCOUNT AND PROFILE

1. Sign in and sign out.
2. Recover a forgotten password.
3. View and edit personal information; the assigned organization is read-only.

### COMPLAINT MANAGEMENT

4. View complaints routed to the representative based on the citizen-selected category.
5. View complaint details, location, attachments, and processing history.
6. Correct a complaint category and reassign it with a recorded reason.
7. View complaint heatmaps and related statistics.
8. Update complaint status and processing notes; attach a resolution photo when resolved.

### GOVERNMENT INITIATIVES

9. Create, update, and delete initiatives within the assigned organization.
10. View written feedback submitted by citizens on the organization's initiatives.

# Actor 3 - System Admin



### ACCOUNT MANAGEMENT

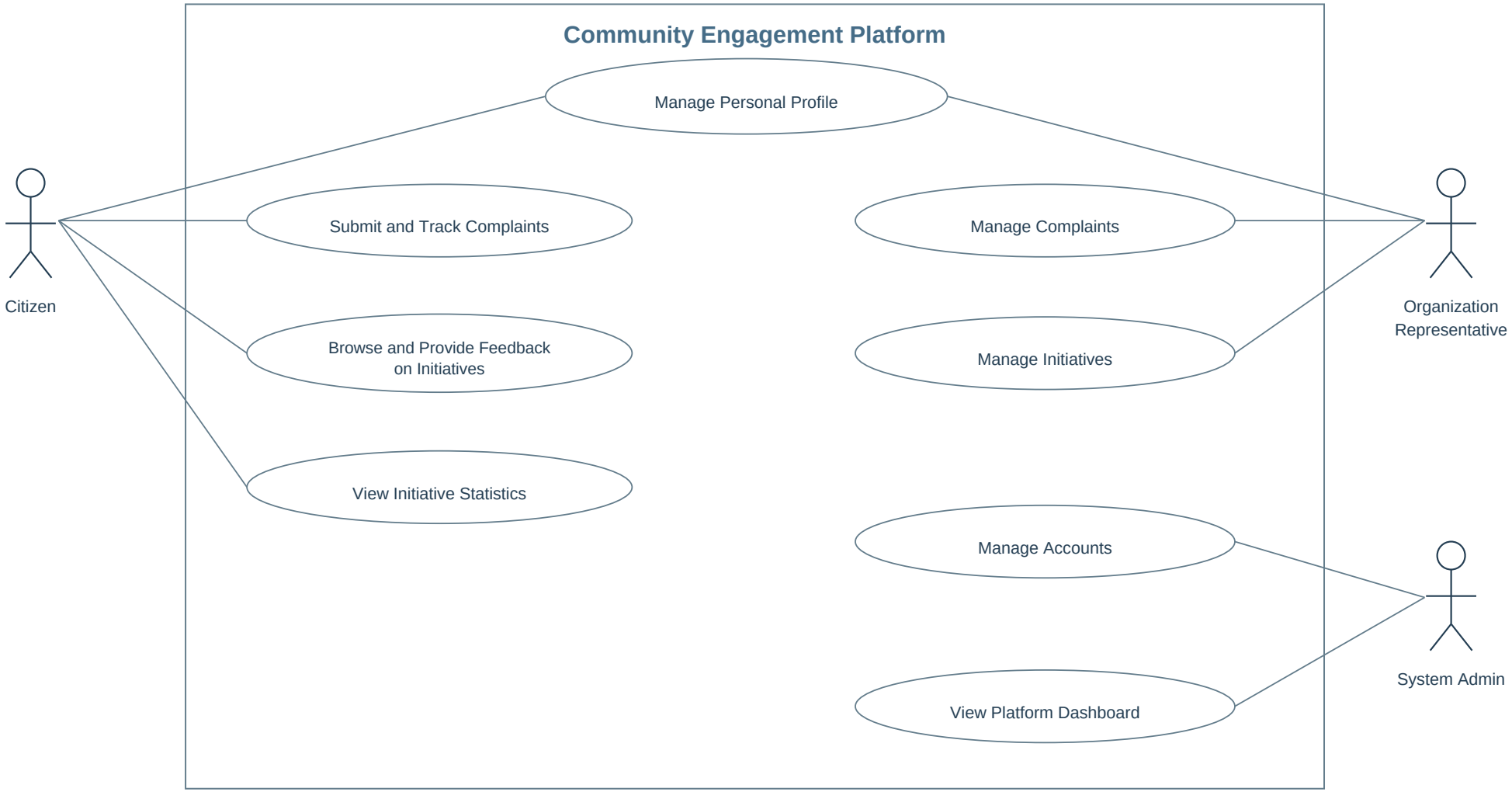
1. Sign in and sign out; create, view, update, activate, and deactivate accounts.
2. Manage Citizen, Organization Representative, and System Admin accounts.
3. Assign an organization and a responsible complaint category to each representative.

### PLATFORM DASHBOARD

4. View aggregate platform dashboards, statistics, and charts for complaints and initiatives.

# High-Level Use Case Diagram






Use cases are carried out according to the permissions of each role.

# Detailed Use Case Specifications

Approved process descriptions, sequence diagrams, and activity diagrams.

**UC 01 Manage Personal Profile** 23

**UC 02 Submit and Track Complaints** 26

**UC 03 Browse and Provide Feedback on Initiatives** 29

**UC 04 View Initiative Statistics** 31

**UC 05 Manage Complaints** 33

**UC 06 Manage Initiatives** 36

**UC 07 Manage Accounts** 39

**UC 08 View Platform Dashboard** 41

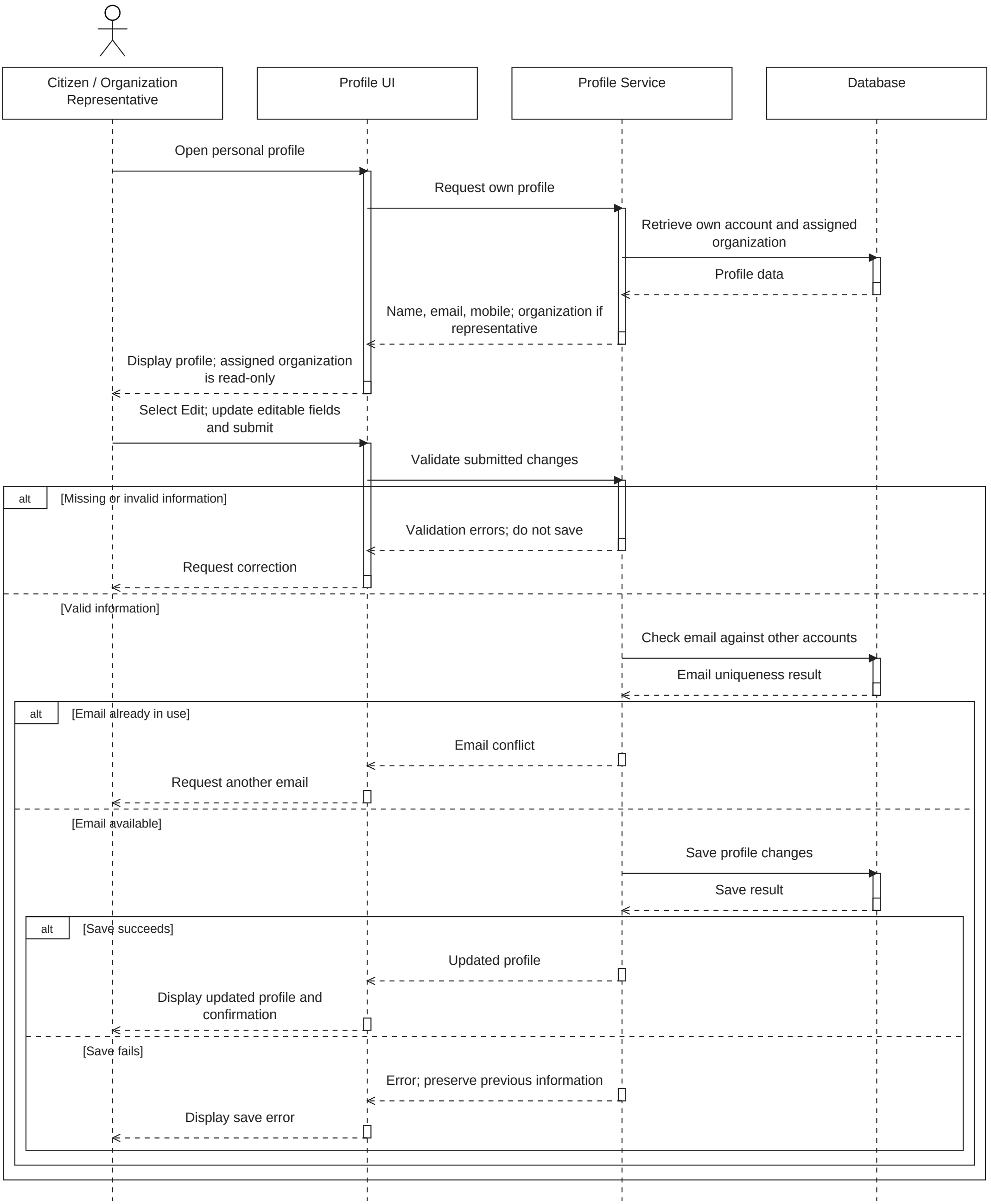
USE CASE 01

# Manage Personal Profile

| Field | Description |
|---|---|
| Use Case Name | Manage Personal Profile |
| Actors | Citizen, Organization Representative |
| Preconditions | The user is signed in. |
| Main Flow | 1. The user opens their personal profile.<br>2. The system displays name, email address, and mobile number. For an Organization Representative, it also displays the assigned organization name as read-only.<br>3. The user selects the edit option.<br>4. The user updates their editable personal information and submits the changes.<br>5. The system validates the submitted information.<br>6. The system saves the changes.<br>7. The system displays the updated profile and a success confirmation. |
| Exception Flows | Exception 1 - Invalid Information: The system identifies missing or invalid information and requests correction without saving.<br>Exception 2 - Email Already in Use: The system requests a different email address.<br>Exception 3 - Save Failure: The system displays an error and retains the previously saved information. |
| Postconditions | Manage Personal Profile completed successfully. |

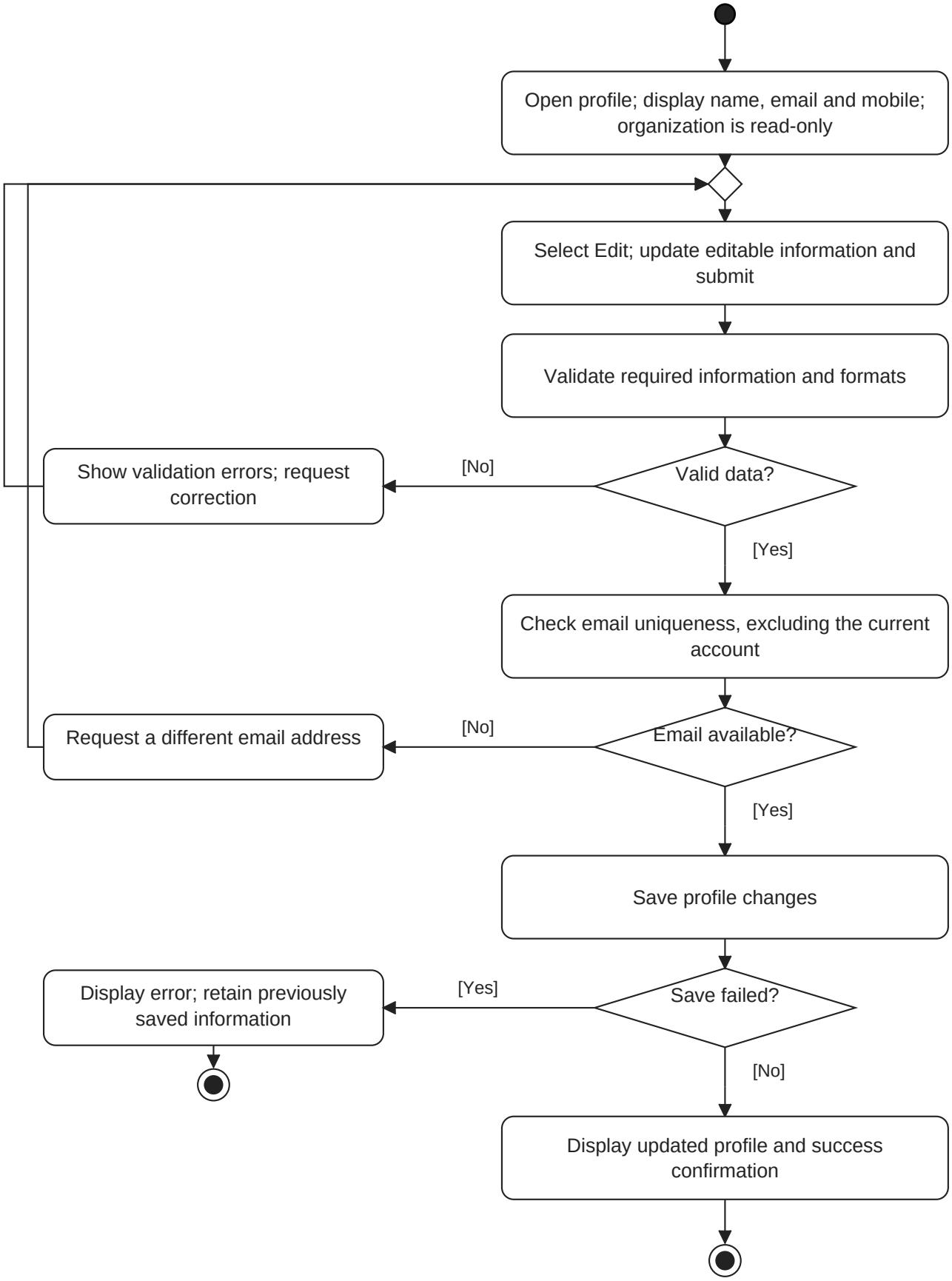
# Sequence diagram (Manage Personal Profile)

Precondition: the user is signed in.



# Manage Personal Profile

## Activity | Manage personal profile



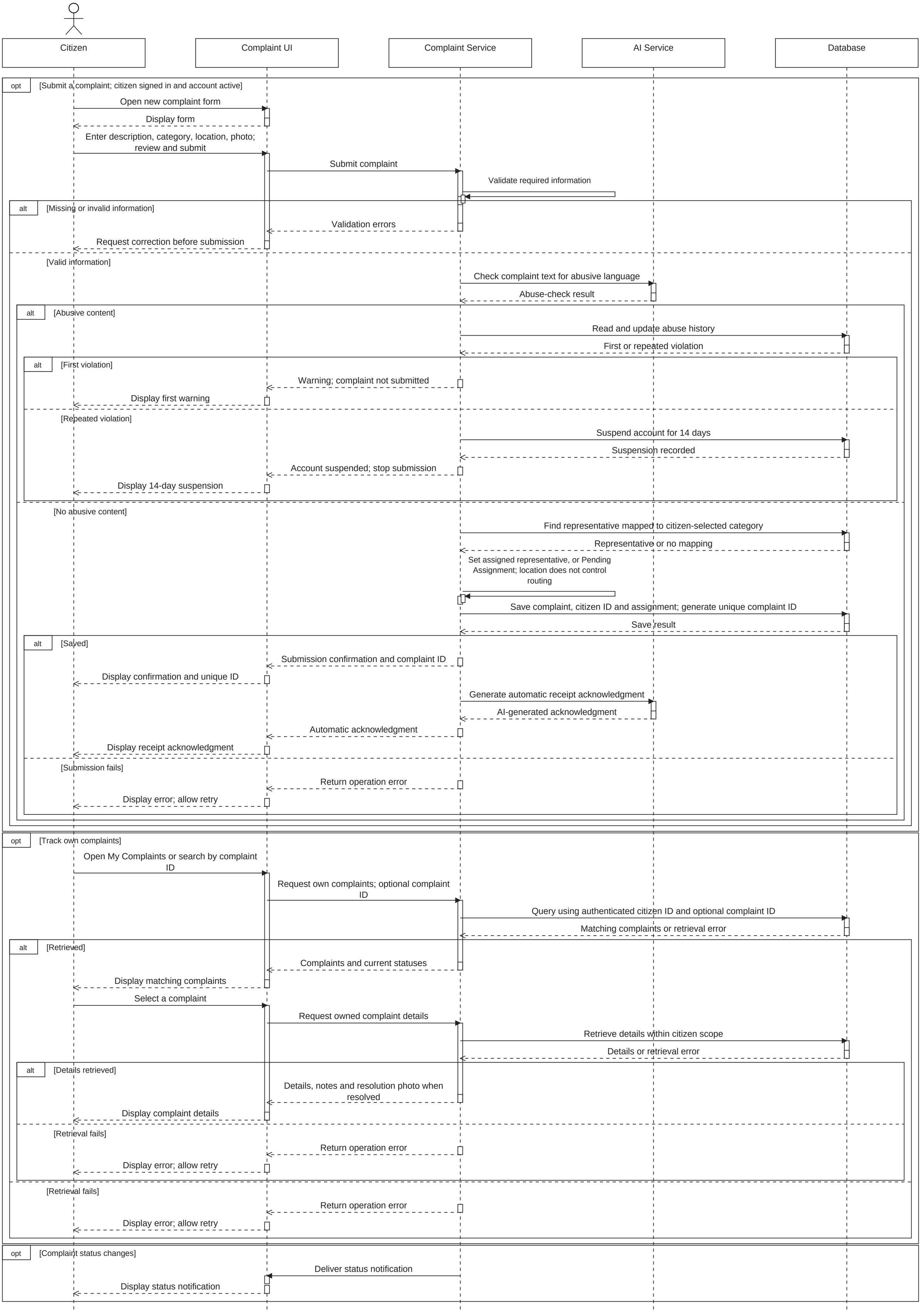
USE CASE 02

# Submit and Track Complaints

| Field | Description |
|---|---|
| **Use Case Name** | Submit and Track Complaints |
| **Actors** | Citizen |
| **Preconditions** | The citizen is signed in and the account is active. |
| **Main Flow** | Submitting a Complaint<br>1. The citizen opens the new complaint form.<br>2. The citizen enters a description, specifies the location, selects a complaint category, and attaches a photo.<br>3. The citizen reviews the information and submits the complaint.<br>4. The system validates the submitted information.<br>5. The system identifies the responsible Organization Representative from the category selected by the citizen, regardless of location.<br>6. The system saves the complaint and assigns it to the responsible representative when one is configured.<br>7. The system displays a submission confirmation and unique complaint ID.<br>8. The AI Virtual Assistant automatically acknowledges receipt of the submitted complaint.<br><br>Tracking Complaints<br>1. The citizen opens My Complaints or searches for a personal complaint using its ID.<br>2. The system displays the relevant complaints and their current statuses.<br>3. The citizen selects a complaint.<br>4. The system displays its details, processing notes, and a resolution photo when resolved.<br>5. The system notifies the citizen when the complaint status changes. |
| **Exception Flows** | Exception 1 - Missing or Invalid Information: The system requests correction before submission.<br>Exception 2 - No Responsible Representative: The system saves the complaint as Pending Assignment until a responsible representative is configured.<br>Exception 3 - Submission or Retrieval Failure: The system displays an error and allows retry.<br>Exception 4 - Abusive Content: The system issues a warning for a first violation and suspends the account for 14 days after repeated abuse. |
| **Postconditions** | Submit and Track Complaints completed successfully. |

# Sequence diagram (Submit and Track Complaints)

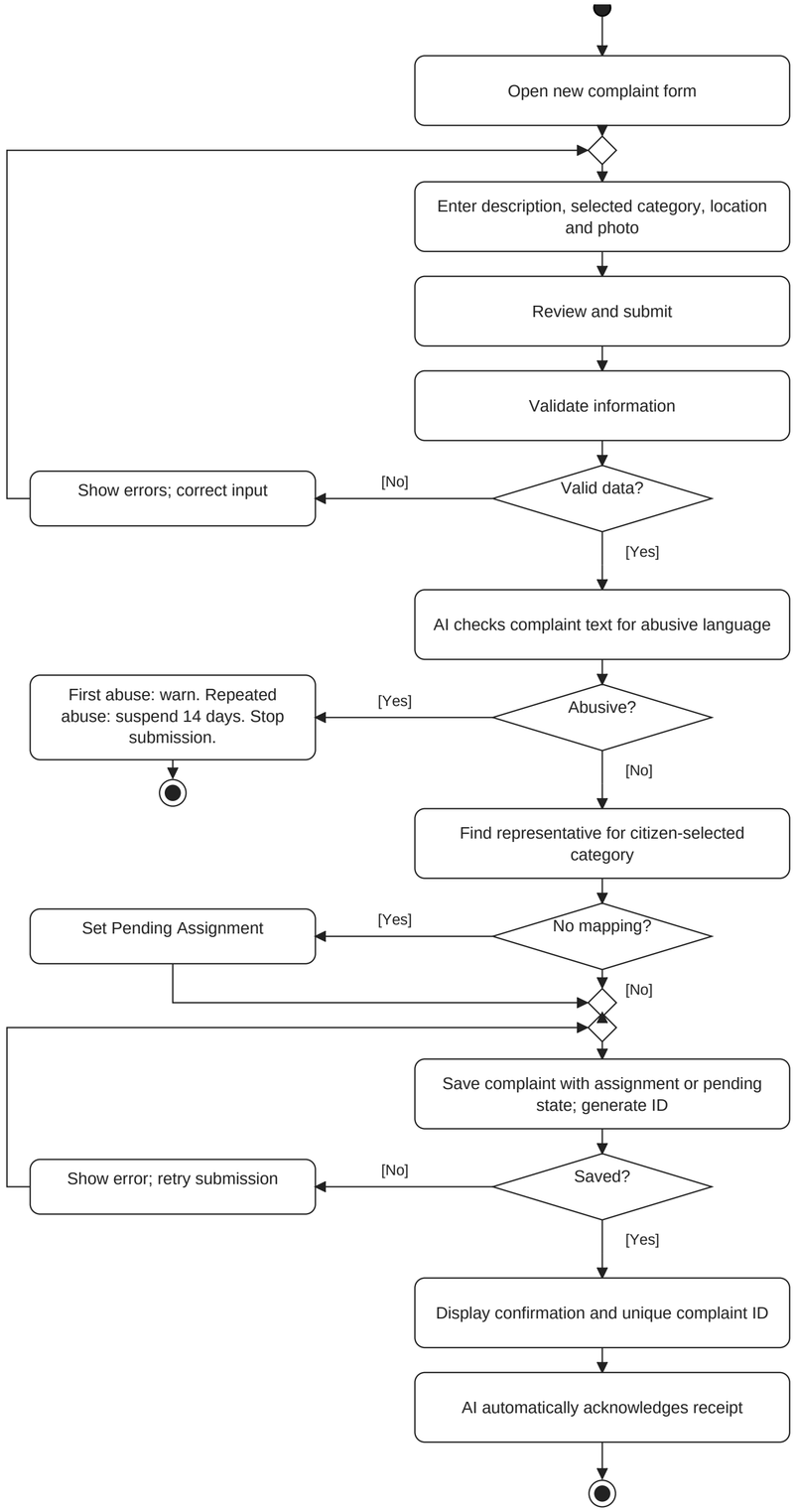
Precondition: the citizen is signed in and the account is active.



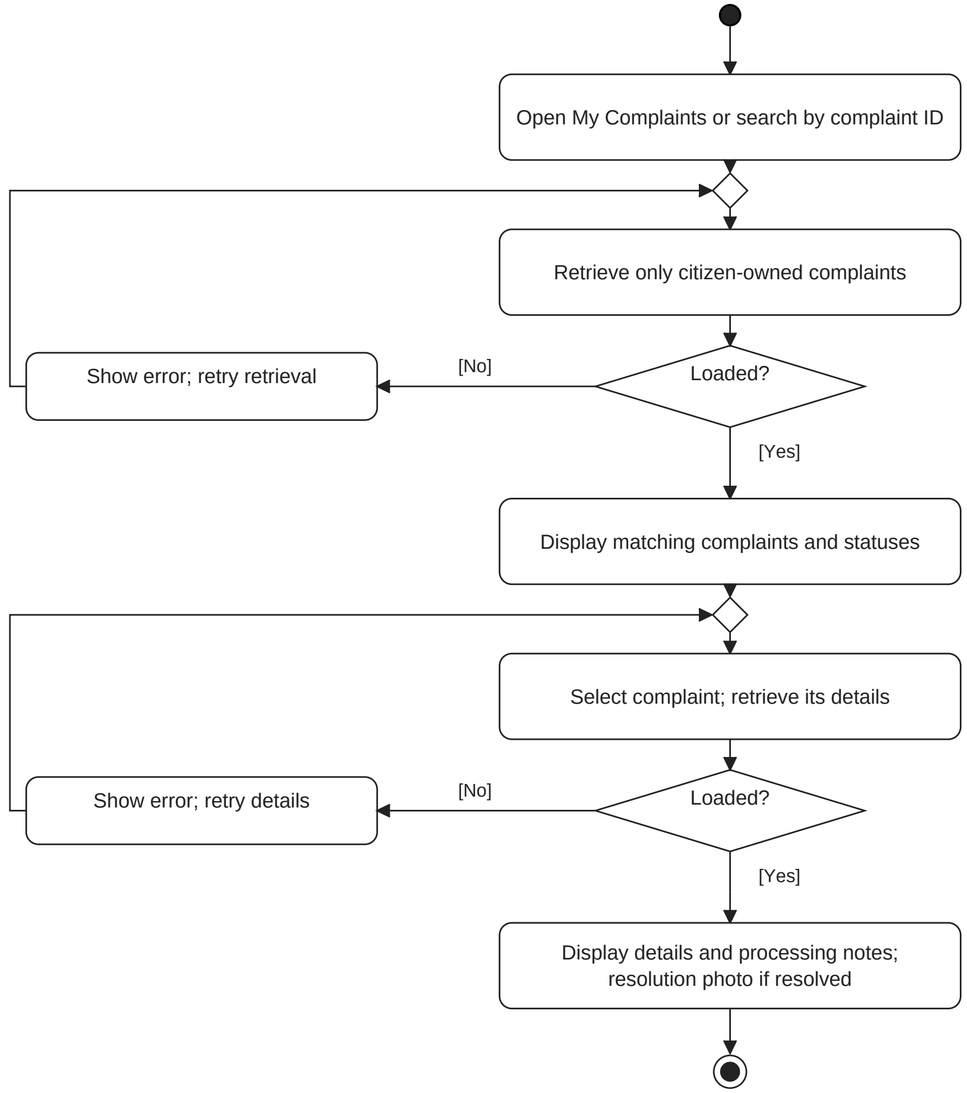
# Activity diagram (Submit and Track Complaints)

Precondition: the citizen is signed in and the account is active.

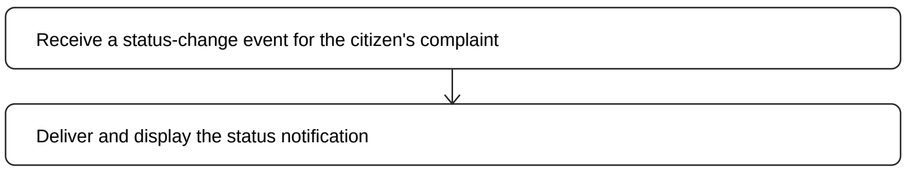
## Submit a complaint




## Track own complaints




### When a complaint status changes



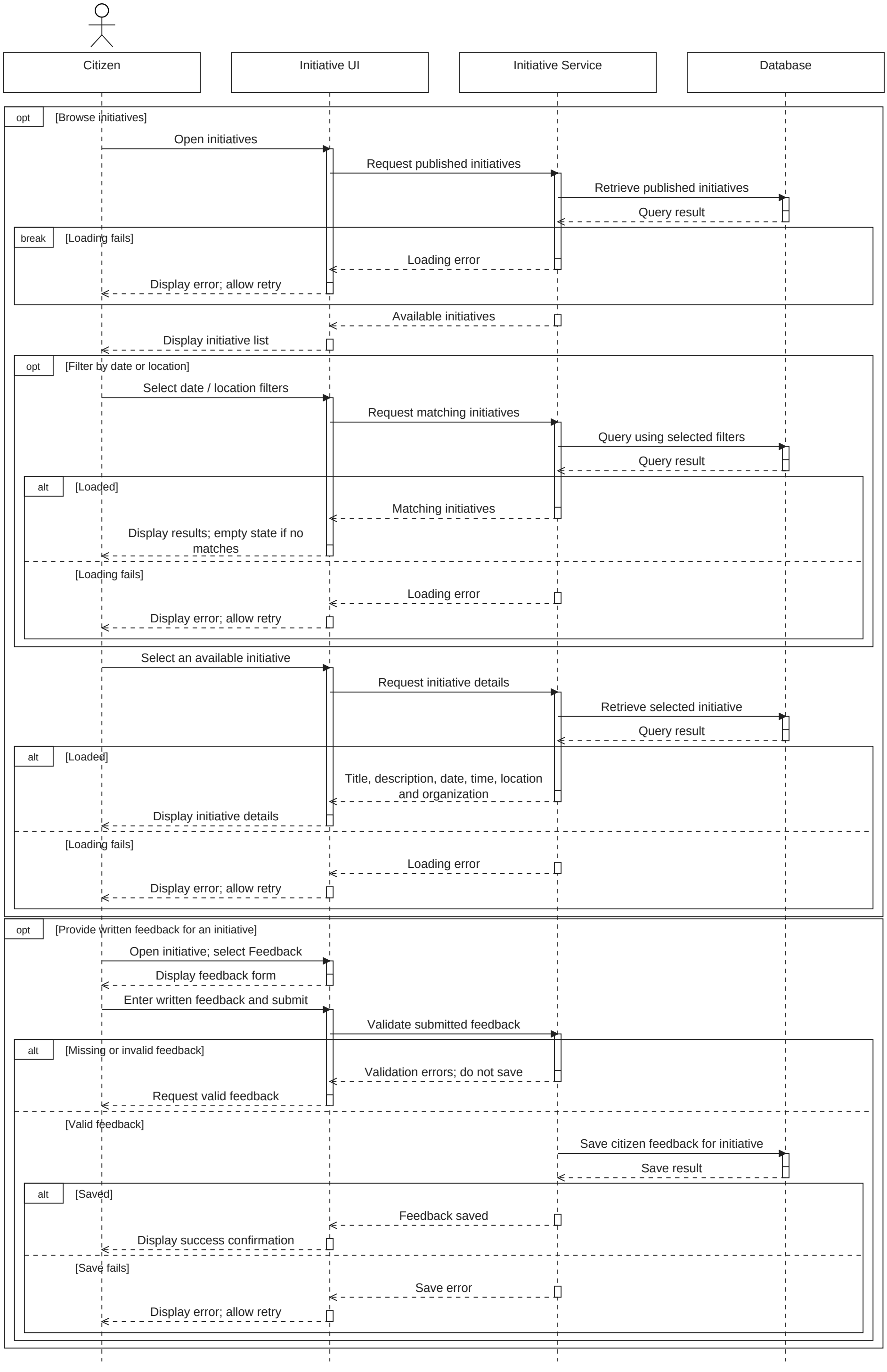
USE CASE 03

# Browse and Provide Feedback on Initiatives

| Field | Description |
|---|---|
| Use Case Name | Browse and Provide Feedback on Initiatives |
| Actors | Citizen |
| Preconditions | The citizen is signed in. |
| Main Flow | Browsing Initiatives<br>1. The citizen opens the initiatives page.<br>2. The system displays available government initiatives.<br>3. The citizen can filter initiatives by date or location.<br>4. The system displays the matching initiatives.<br>5. The citizen selects an initiative.<br>6. The system displays its title, description, date, time, location, and organizing organization.<br><br>Providing Feedback<br>1. The citizen opens an initiative and selects the feedback option.<br>2. The system displays the feedback form.<br>3. The citizen enters written feedback and submits it.<br>4. The system validates and saves the feedback.<br>5. The system displays a success confirmation. |
| Exception Flows | Exception 1 - Missing or Invalid Feedback: The system requests valid feedback before saving. |
| | Exception 2 - Loading or Saving Failure: The system displays an error and allows retry. |
| Postconditions | Browse and Provide Feedback on Initiatives completed successfully. |

# Sequence diagram (Browse and Provide Feedback)

Precondition: the citizen is signed in.



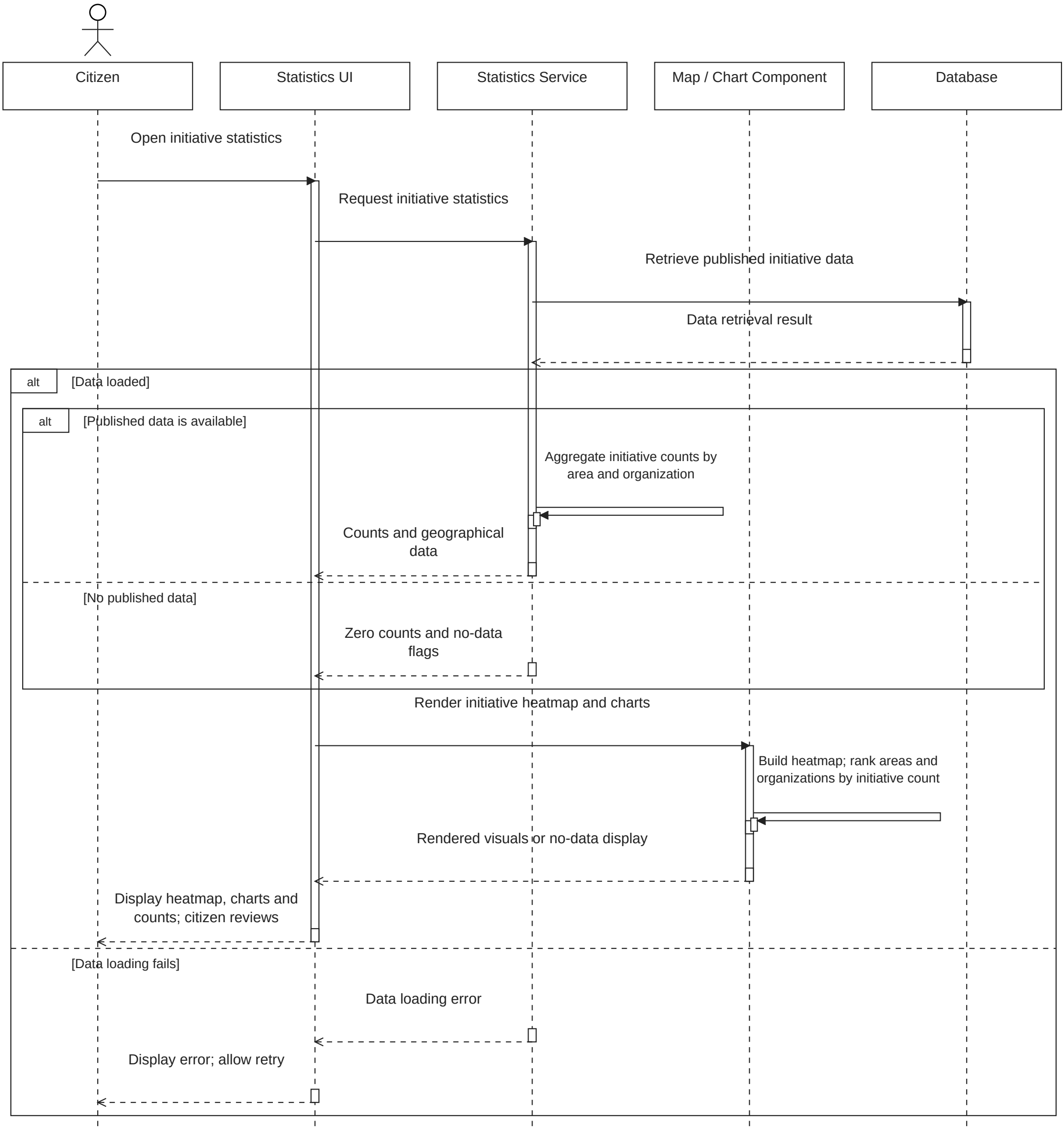
USE CASE 04

# View Initiative Statistics

| Field | Description |
|---|---|
| Use Case Name | View Initiative Statistics |
| Actors | Citizen |
| Preconditions | The citizen is signed in. |
| Main Flow | 1. The citizen opens the initiative statistics page.<br>2. The system retrieves published initiative data.<br>3. The system displays a geographical heatmap of initiatives.<br>4. The system displays charts showing the areas with the most initiatives and the organizations publishing the most initiatives.<br>5. The citizen reviews the information. |
| Exception Flows | Exception 1 - No Available Data: The dashboard displays zero counts or a no-data message. |
| | Exception 2 - Data Loading Failure: The system displays an error and allows retry. |
| Postconditions | View Initiative Statistics completed successfully. |

# Sequence diagram (View Initiative Statistics)

Precondition: the citizen is signed in.



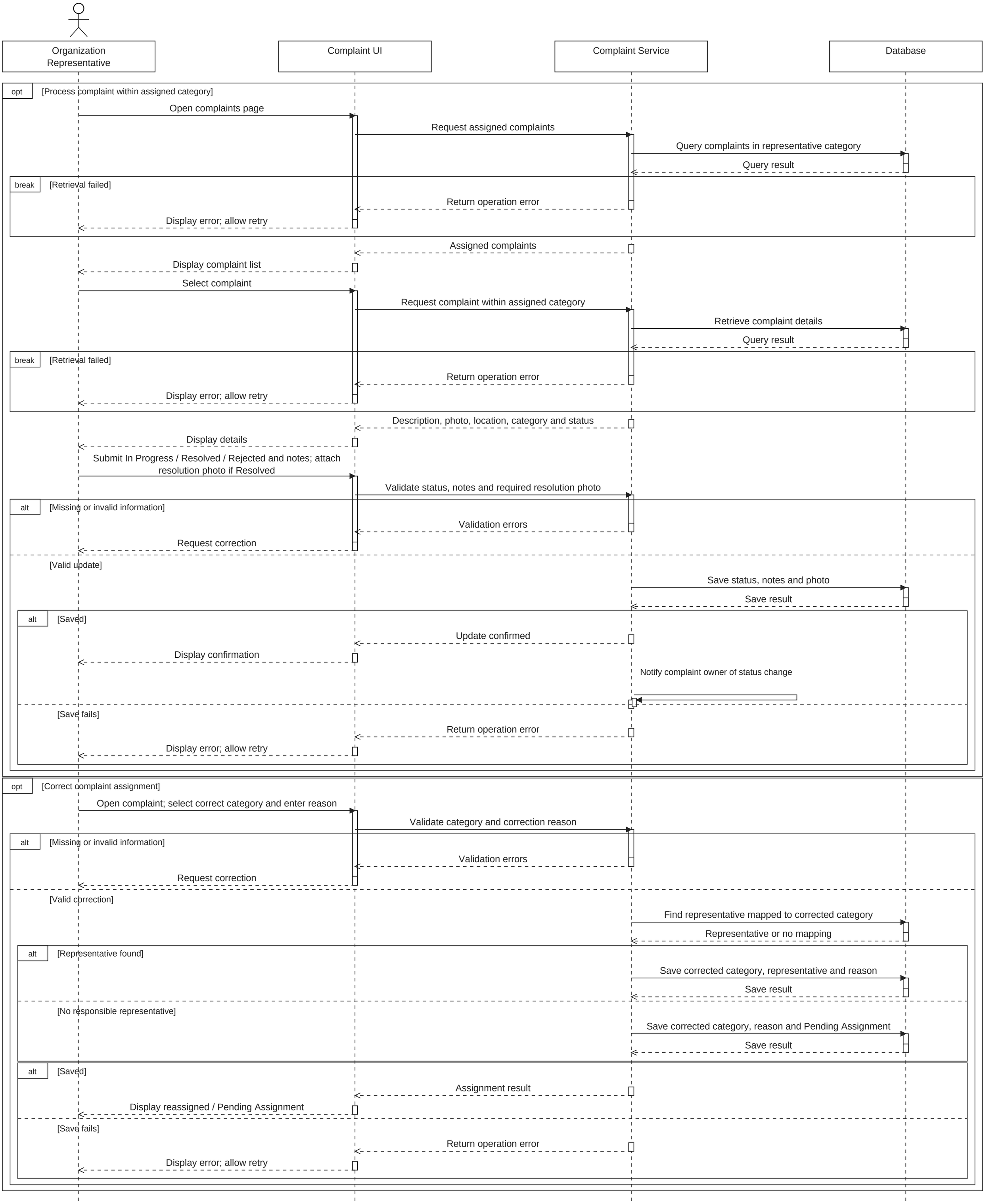
USE CASE 05

# Manage Complaints

| Field | Description |
|---|---|
| Use Case Name | Manage Complaints |
| Actors | Organization Representative |
| Preconditions | The representative is signed in. Processing or reassignment requires at least one complaint within the assigned category. |
| Main Flow | Processing Complaints<br>1. The representative opens the complaints page.<br>2. The system displays complaints assigned to the representative's complaint category.<br>3. The representative selects a complaint.<br>4. The system displays its description, photo, location, category, and status.<br>5. The representative updates the status to In Progress, Resolved, or Rejected and adds notes. A resolution photo is required when resolved.<br>6. The system validates and saves the changes.<br>7. The system confirms the update and notifies the citizen of the status change.<br><br>Correcting Complaint Assignment<br>1. The representative opens a complaint requiring correction.<br>2. The representative selects the appropriate category and provides a reason.<br>3. The system identifies the representative responsible for that category.<br>4. The system saves the corrected category and assignment, and records the reason. |
| Exception Flows | Exception 1 - Missing or Invalid Information: The system requests missing notes, correction reason, or mandatory resolution photo. |
| | Exception 2 - No Responsible Representative: The complaint remains Pending Assignment until a valid category mapping is available. |
| | Exception 3 - Loading or Saving Failure: The system displays an error and allows retry. |
| Postconditions | Manage Complaints completed successfully. |

# Sequence diagram (Manage Complaints)

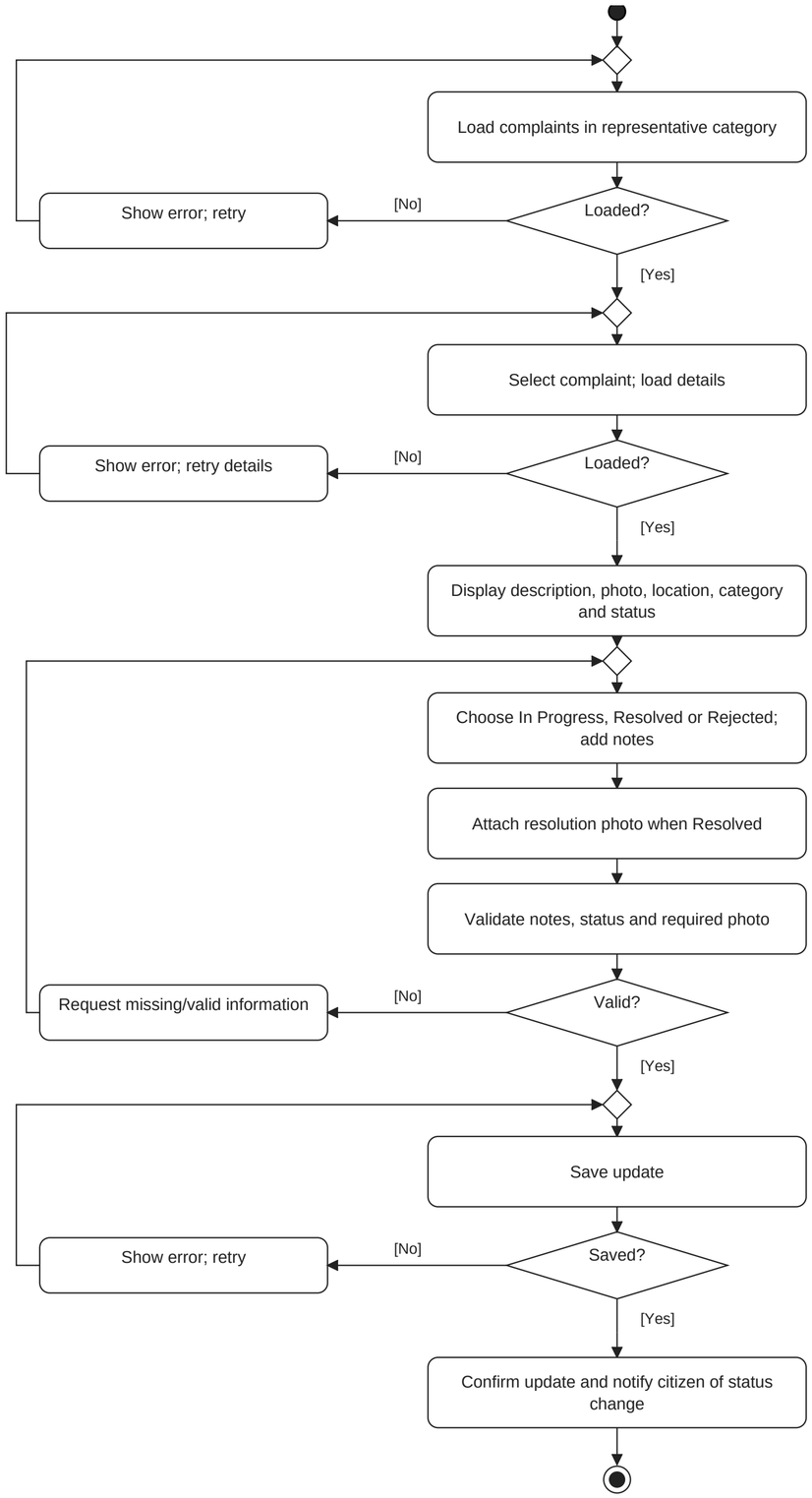
Precondition: the representative is signed in; at least one complaint exists within the assigned category.



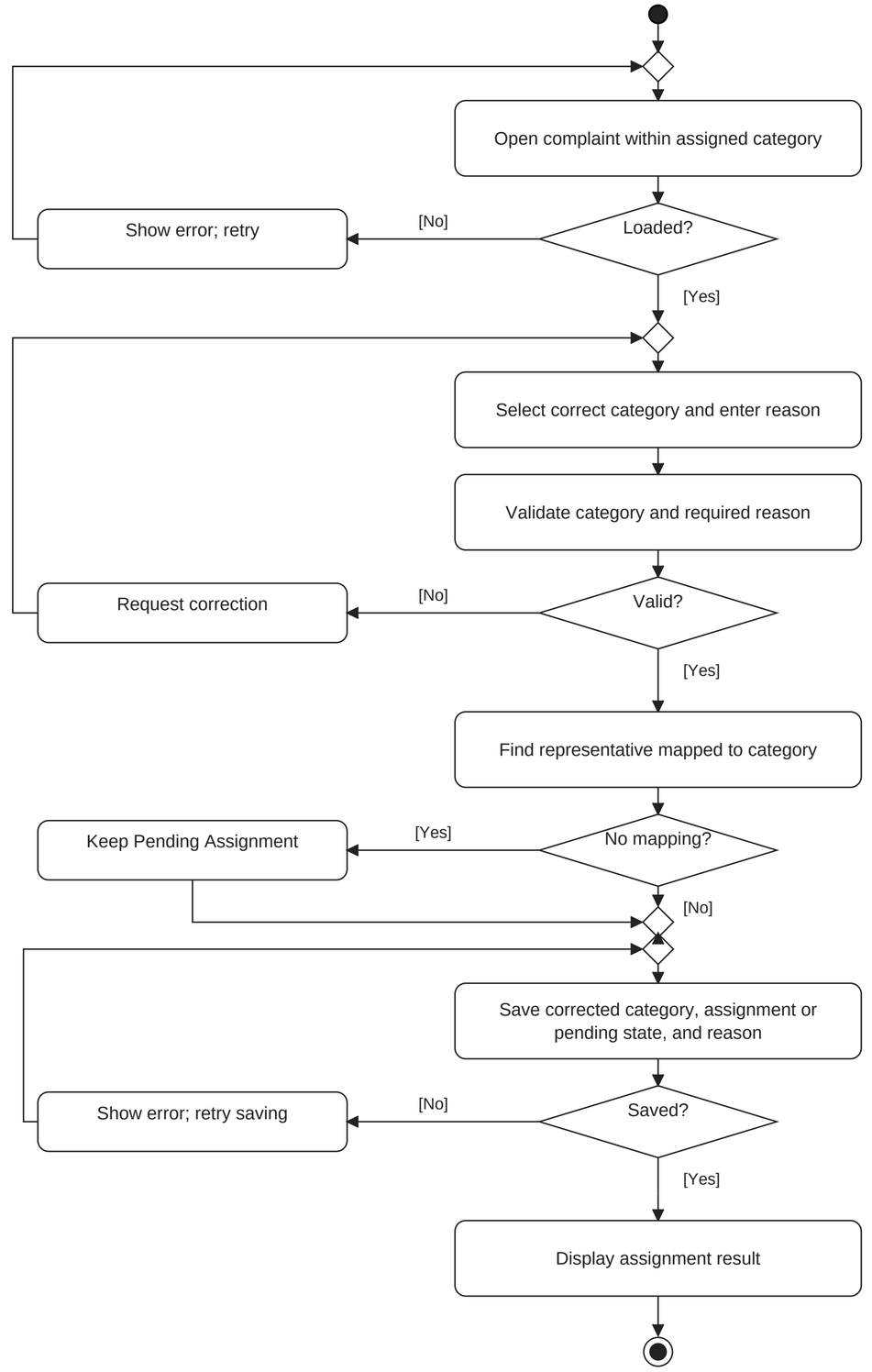
# Activity diagram (Manage Complaints)

Precondition: the representative is signed in; at least one complaint exists within the assigned category.

## Process assigned complaints




## Correct complaint assignment



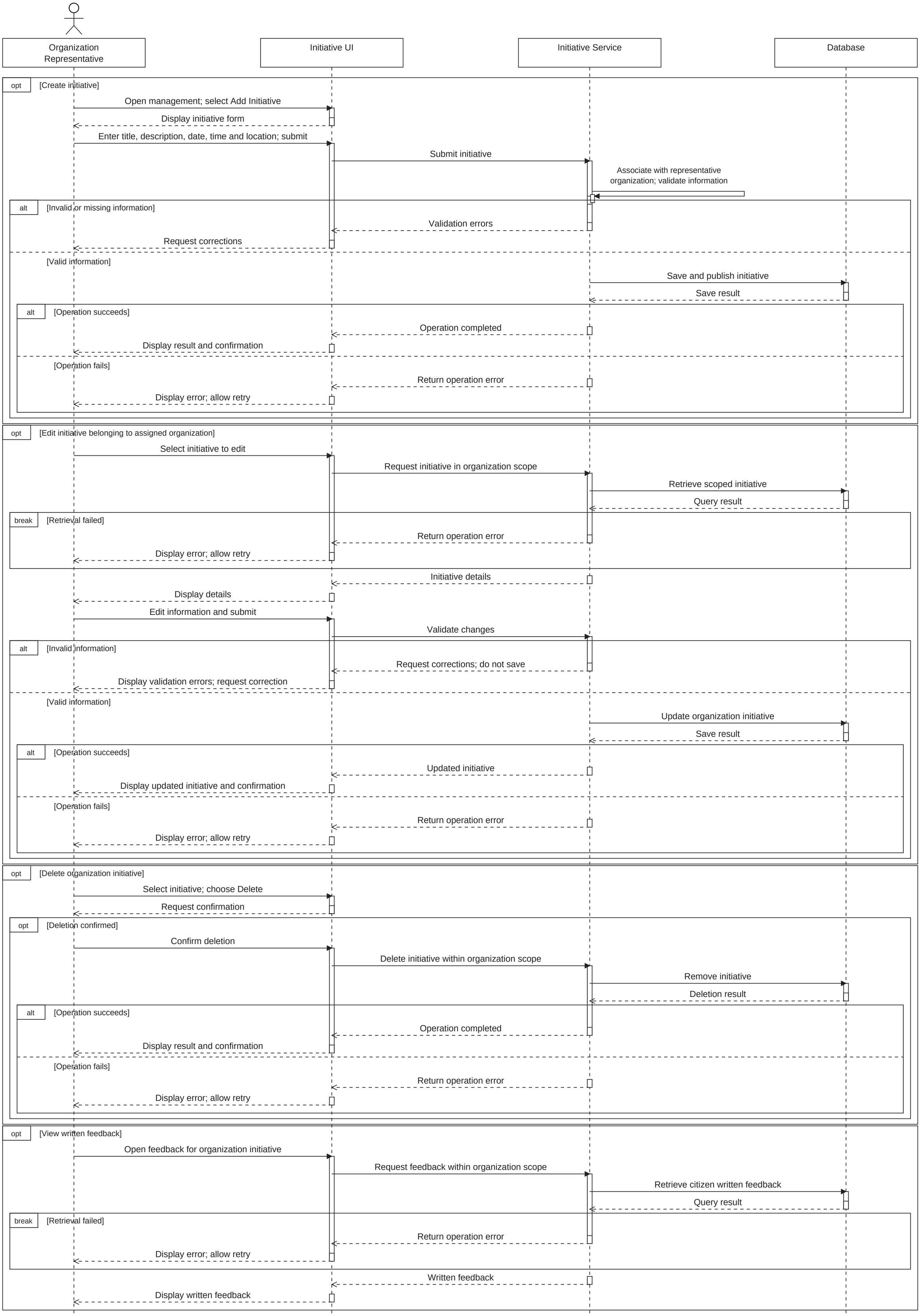
USE CASE 06

# Manage Initiatives

| Field | Description |
|---|---|
| Use Case Name | Manage Initiatives |
| Actors | Organization Representative |
| Preconditions | The representative is signed in. Editing or deleting requires an initiative belonging to the assigned organization. |
| Main Flow | Creating an Initiative<br>1. The representative opens initiative management and selects Add Initiative.<br>2. The system displays the initiative form.<br>3. The representative enters the title, description, date, time, and location.<br>4. The system associates the initiative with the representative's organization.<br>5. The representative submits the information.<br>6. The system validates, saves, and publishes the initiative.<br>7. The system displays a success confirmation.<br><br>Editing an Initiative<br>1. The representative selects an initiative from the assigned organization.<br>2. The system displays its details.<br>3. The representative edits and submits the changes.<br>4. The system validates and saves them.<br>5. The system displays the updated initiative.<br><br>Deleting an Initiative<br>1. The representative selects an initiative and chooses Delete.<br>2. The system requests confirmation.<br>3. The representative confirms deletion.<br>4. The system removes the initiative and displays confirmation.<br><br>Viewing Feedback<br>1. The representative opens feedback submitted for an organization initiative.<br>2. The system displays the citizens' written feedback. |
| Exception Flows | Exception 1 - Missing or Invalid Information: The system requests corrections before saving. |
| | Exception 2 - Operation Failure: The system shows an error and allows retry. |
| Postconditions | Manage Initiatives completed successfully. |

# Sequence diagram (Manage Initiatives)

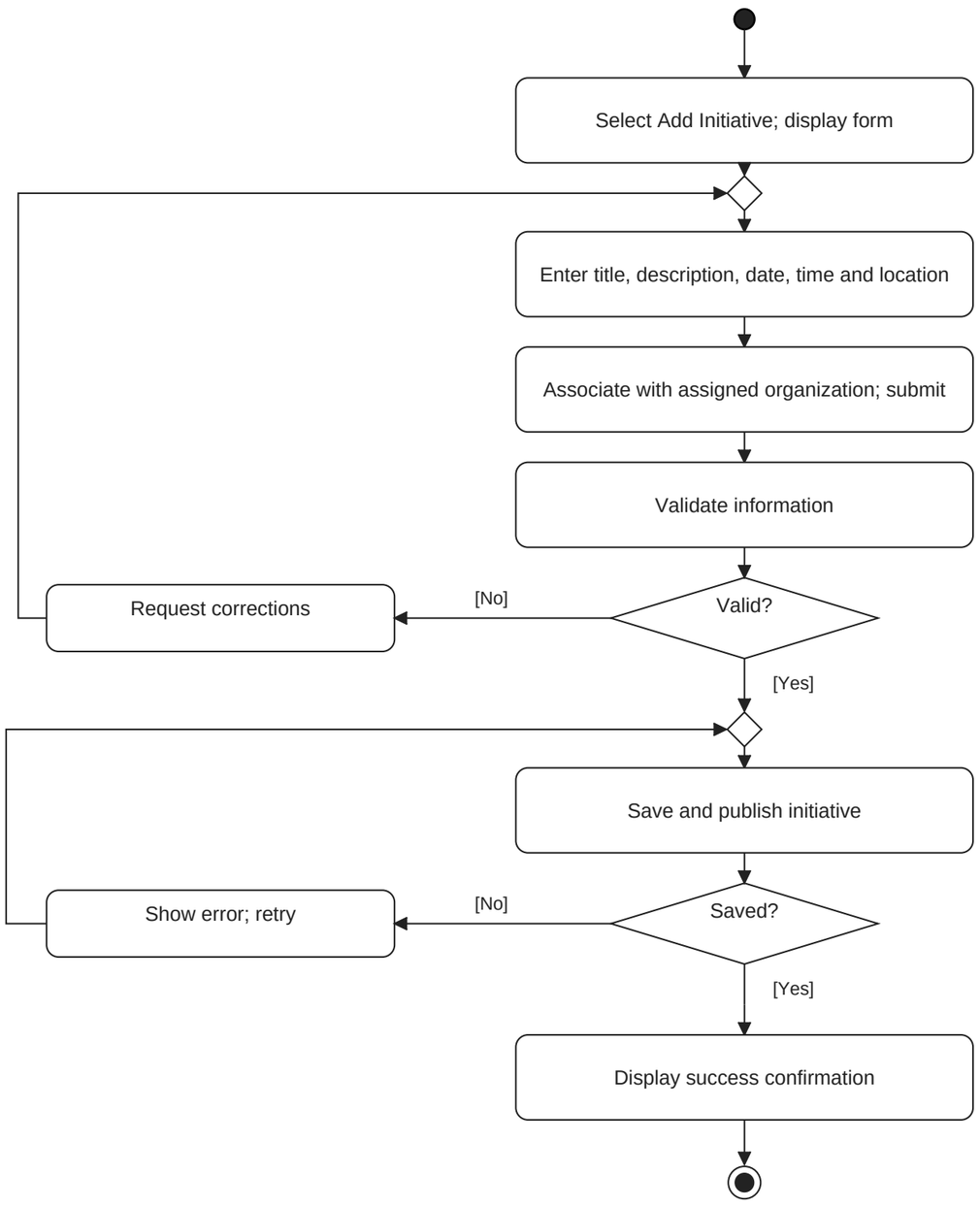
Precondition: the representative is signed in; edit/delete require an initiative in the assigned organization.



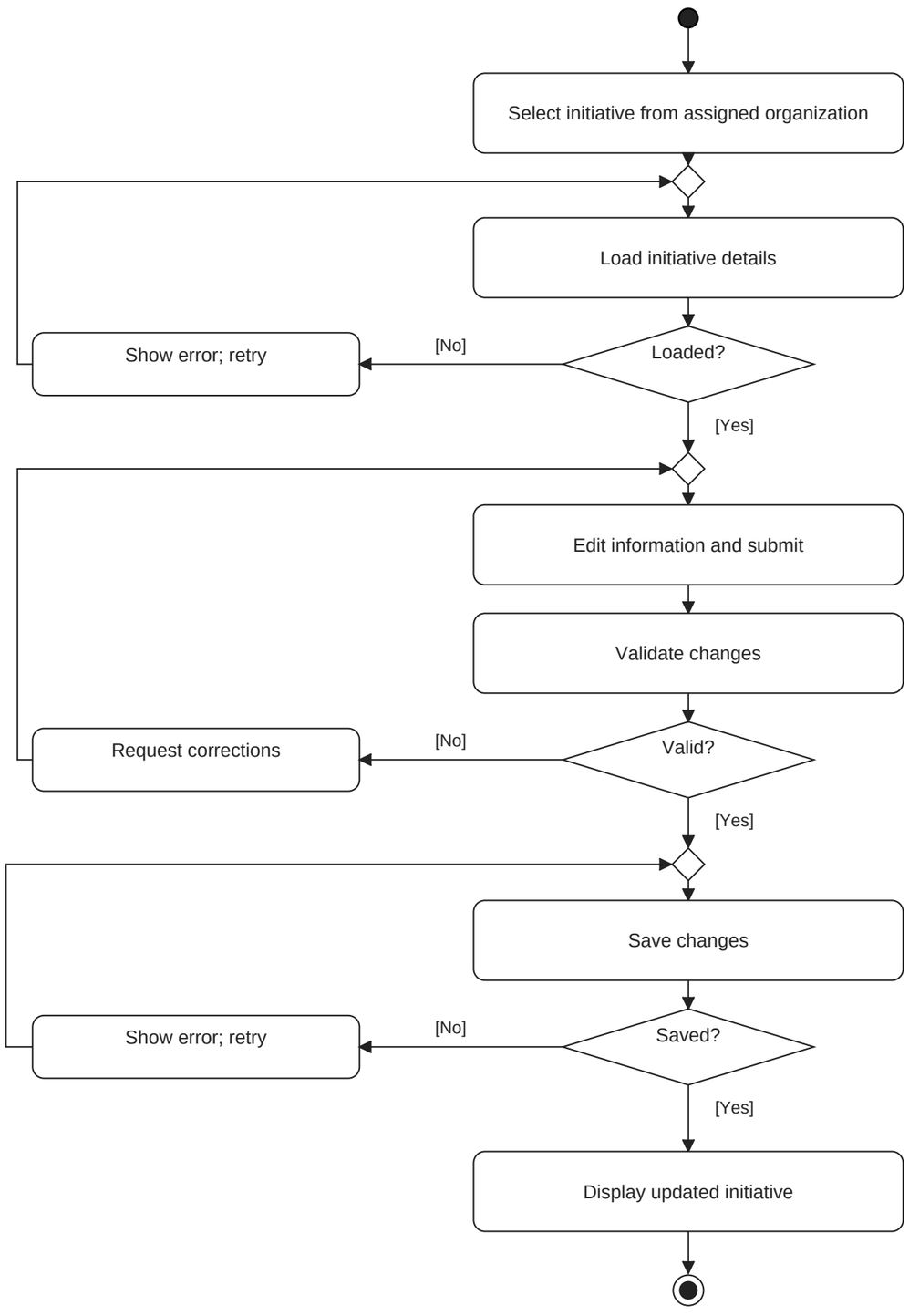
# Activity diagram (Manage Initiatives)

Precondition: the representative is signed in; edit/delete require an initiative in the assigned organization.

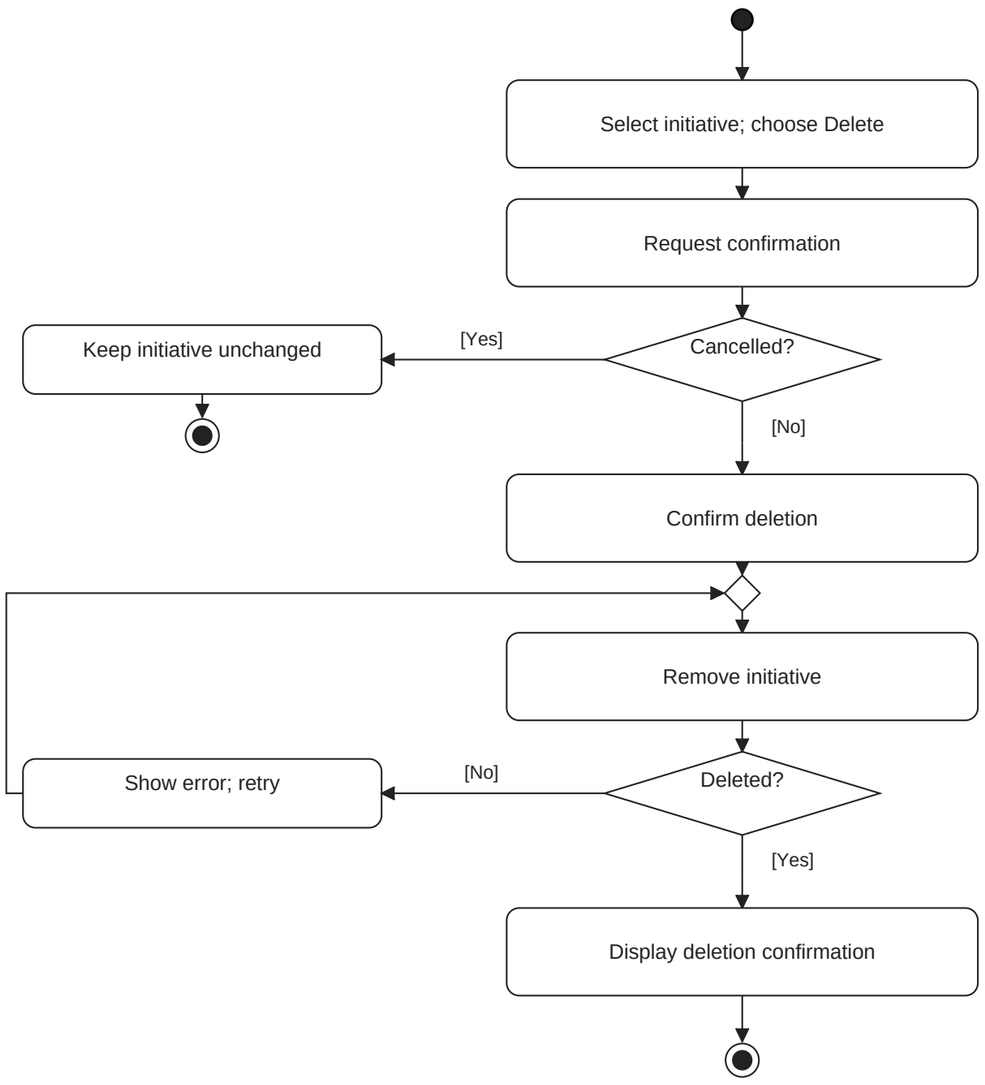
## Create initiative



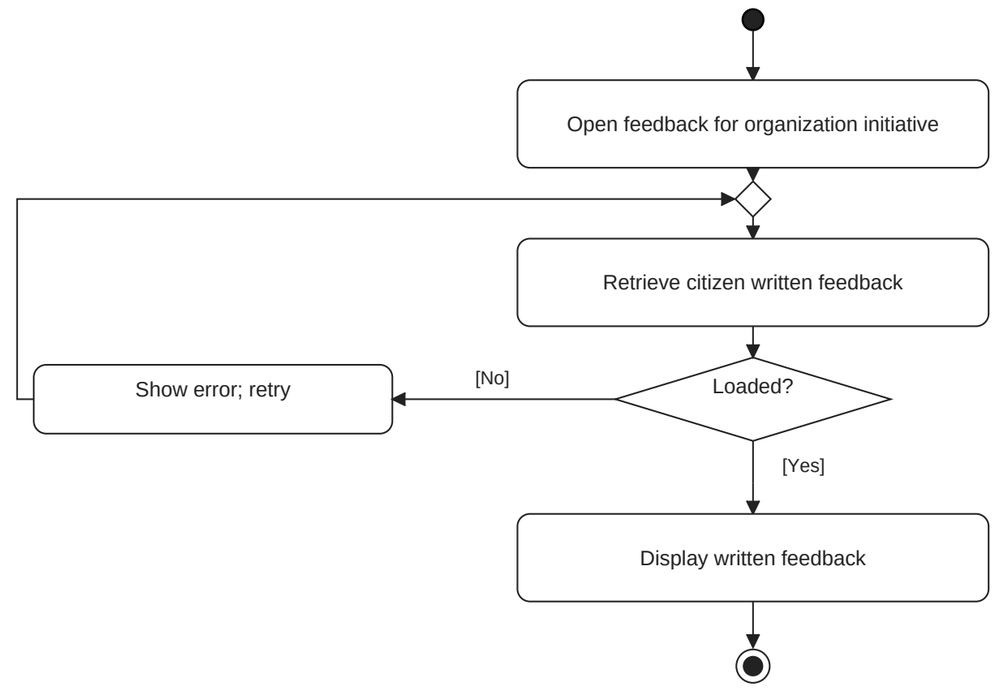

## Edit organization initiative




## Delete organization initiative




## View written feedback



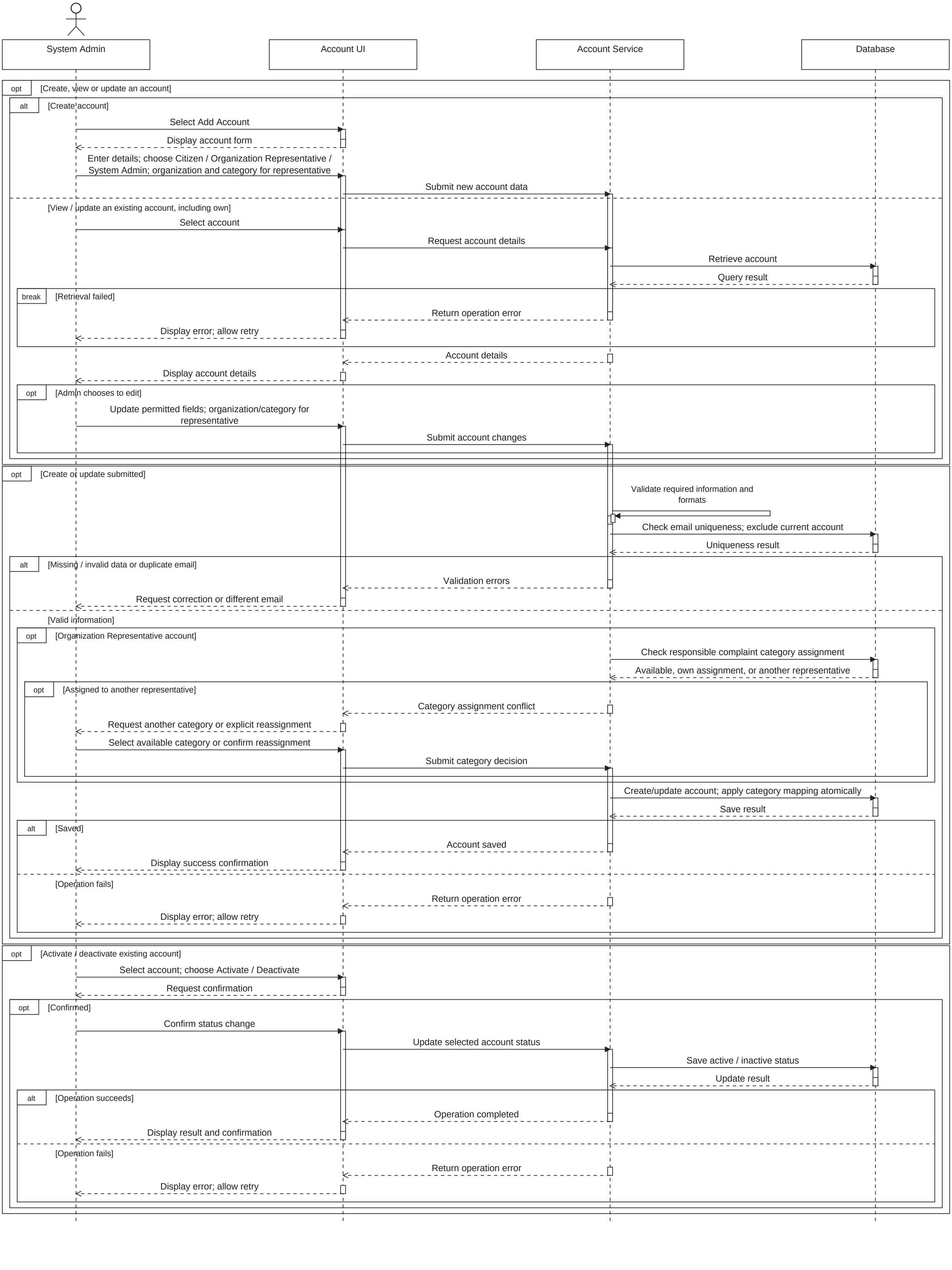
USE CASE 07

# Manage Accounts

| Field | Description |
|---|---|
| Use Case Name | Manage Accounts |
| Actors | System Admin |
| Preconditions | The admin is signed in. Viewing, updating, or changing account status requires an existing account. |
| Main Flow | Creating an Account<br>1. The admin opens account management and selects Add Account.<br>2. The system displays the account form.<br>3. The admin enters account details and selects Citizen, Organization Representative, or System Admin.<br>4. For a representative, the admin specifies the organization and responsible complaint category.<br>5. The admin submits the information.<br>6. The system validates and creates the account.<br>7. The system displays a success confirmation.<br><br>Viewing and Updating Accounts<br>1. The admin selects an account, including their own.<br>2. The system displays account details.<br>3. The admin updates permitted information, including organization/category for representatives.<br>4. The system validates and saves changes.<br>5. The system displays confirmation.<br><br>Activating or Deactivating Accounts<br>1. The admin selects an account and chooses Activate or Deactivate.<br>2. The system asks for confirmation.<br>3. The admin confirms the operation.<br>4. The system updates account status and displays confirmation. |
| Exception Flows | Exception 1 - Missing, Invalid, or Duplicate Information: The system requests correction or a different email address. |
| | Exception 2 - Category Already Assigned: The admin must select another category or explicitly reassign responsibility. |
| | Exception 3 - Operation Failure: The system displays an error and allows retry. |
| Postconditions | Manage Accounts completed successfully. |

# Sequence diagram (Manage Accounts)

Precondition: the admin is signed in; view, update and status changes require an existing account.



USE CASE 08

# View Platform Dashboard

| Field | Description |
|---|---|
| Use Case Name | View Platform Dashboard |
| Actors | System Admin |
| Preconditions | The admin is signed in. |
| Main Flow | 1. The admin opens the platform dashboard.<br>2. The system retrieves complaint and initiative data.<br>3. The system displays aggregate complaint statistics and distribution by status and category.<br>4. The system displays initiative statistics and available feedback summaries.<br>5. The system displays relevant geographic heatmaps and charts.<br>6. The admin reviews the dashboard. |
| Exception Flows | Exception 1 - No Available Data: The system displays zero counts or no-data messages, and an empty heatmap when appropriate. |
| | Exception 2 - Data Loading or Calculation Failure: The system displays an error and allows retry. |
| Postconditions | View Platform Dashboard completed successfully. |

# Sequence diagram (View Platform Dashboard)

Precondition: the admin is signed in.

